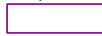
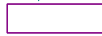


Daisy



File: Seed3-DevKit-Desktop_1.kicad_sch

Peripherals



File: Seed3-DevKit-Desktop_2.kicad_sch

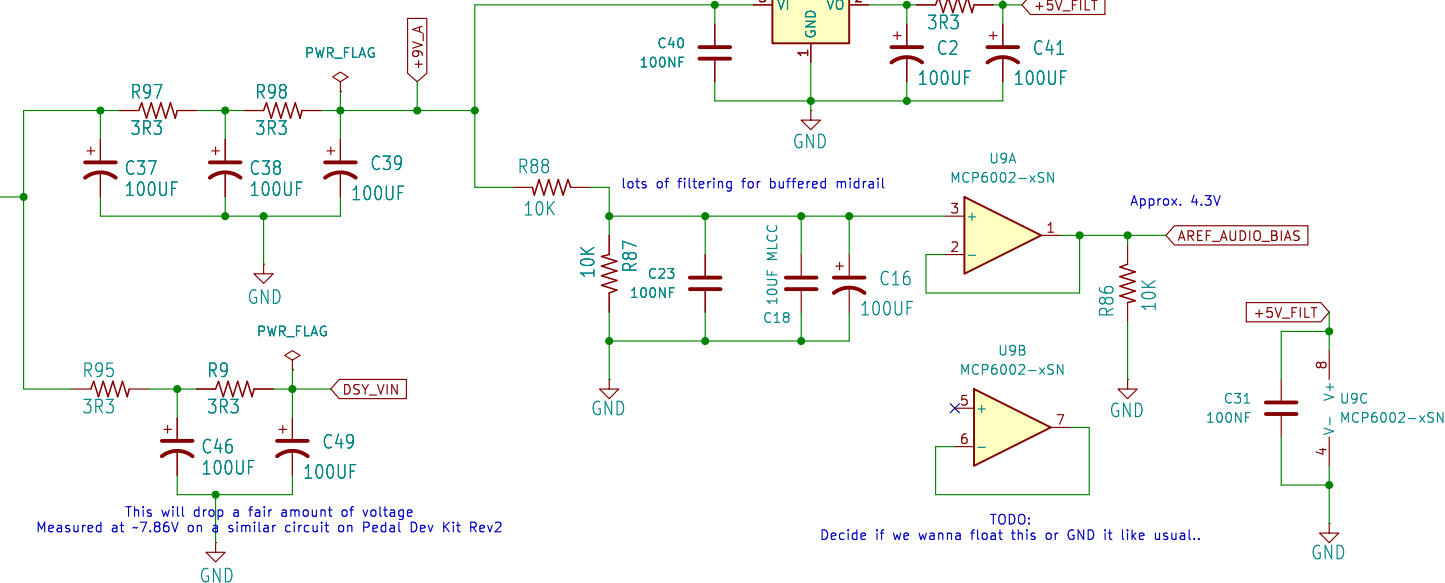
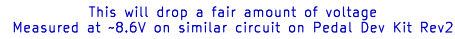
Audio



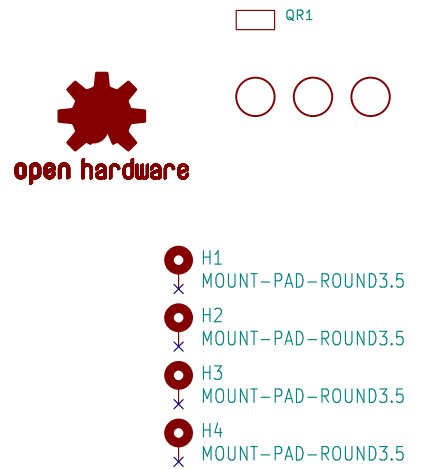
File: audio.kicad_sch

- Headphone Amp
- DAC output op-amp
- Reference voltage for CV inputs
- 6002 buffer for AREF_AUDIO_BIAS

- Power headphone amp from +5V_{SA}
- Change DAC output op-amp to a rail-to-rail opamp that supports higher voltage (i.e. OPA1652)
- Power DAC output op-amp from +9V_A
- Add LM4040 for +5V reference for CV inputs
- Use TL074 or something for AREF_AUDIO_BIAS buffer instead, and power at +9V

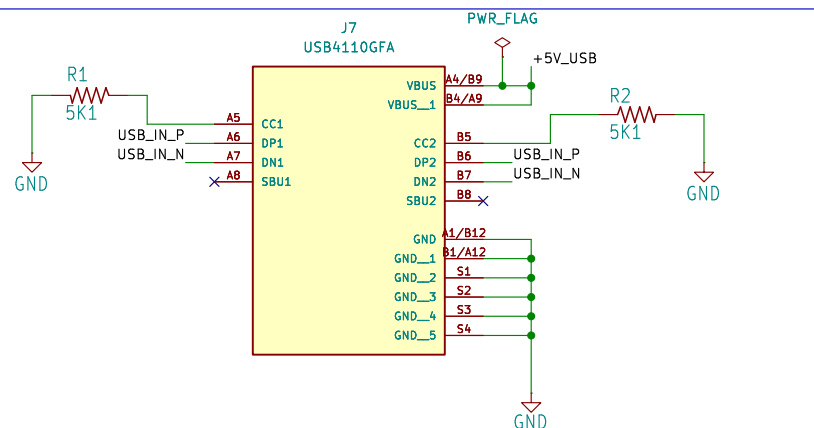


Decide if we wanna float this or GND it like usual..



Pin configuration for U14 (USB to UART bridge):

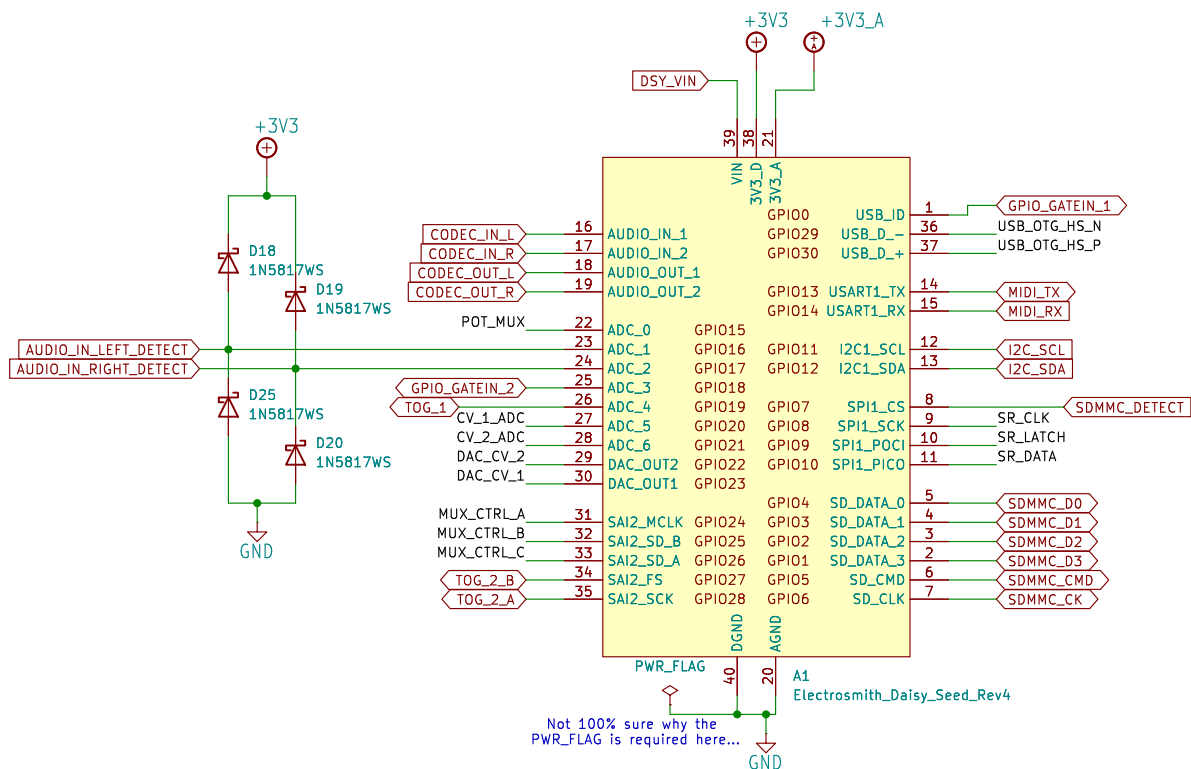
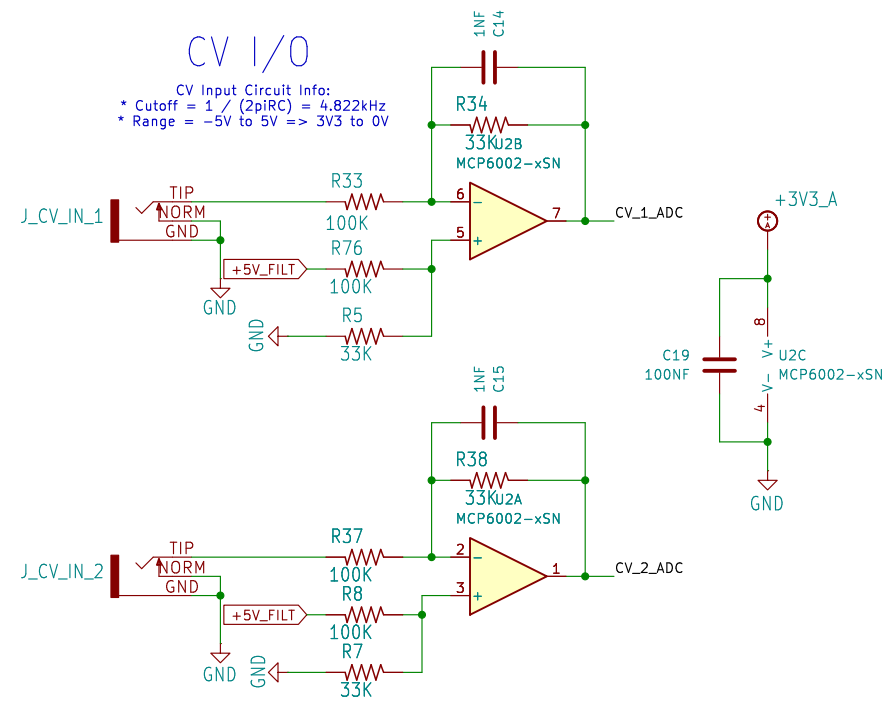
- Pin 1: USB_IN_P (connected to GND)
- Pin 2: GND (connected to GND)
- Pin 3: USB_IN_N (connected to GND)
- Pin 4: I/O2 (connected to USB_OTG_HS_N)
- Pin 5: +5V_USB (connected to +5V_USB)
- Pin 6: I/O1 (connected to USB_OTG_HS_P)



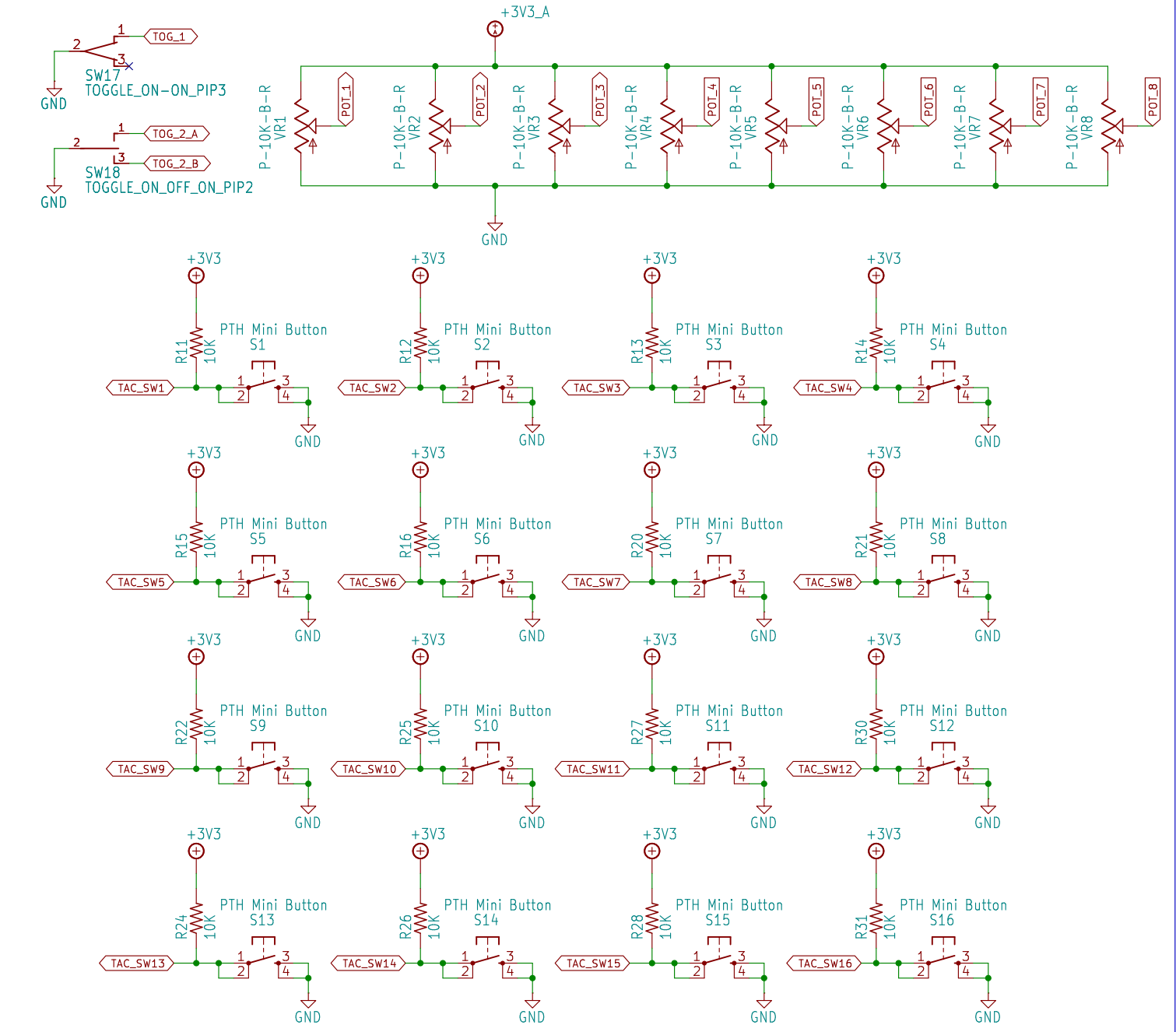
DAC / Gate Output

The circuit diagram shows two op-amp buffers, U6A and U6B, both MCP6002-xSN. U6A is configured with a non-inverting input (pin 3) connected to DAC_CV_1 and a voltage divider consisting of R48 (120K) and R49 (62K). The output (pin 1) is connected to a 100R resistor (R55) and then to a TIP NORM GND connector labeled J_CV_OUT_1. U6B is configured with a non-inverting input (pin 5) connected to DAC_CV_2 and a voltage divider consisting of R56 (120K) and R57 (62K). The output (pin 7) is connected to a 100R resistor (R60) and then to a TIP NORM GND connector labeled J_CV_OUT_2. A separate voltage divider is shown at the top right, consisting of a 5V_FILT source connected to a 100NF capacitor (C20) and a voltage divider (U6C, MCP6002-xSN) with pins 8 and 4 connected to GND. The output of this divider is connected to a TIP NORM GND connector labeled J_CV_OUT_1.

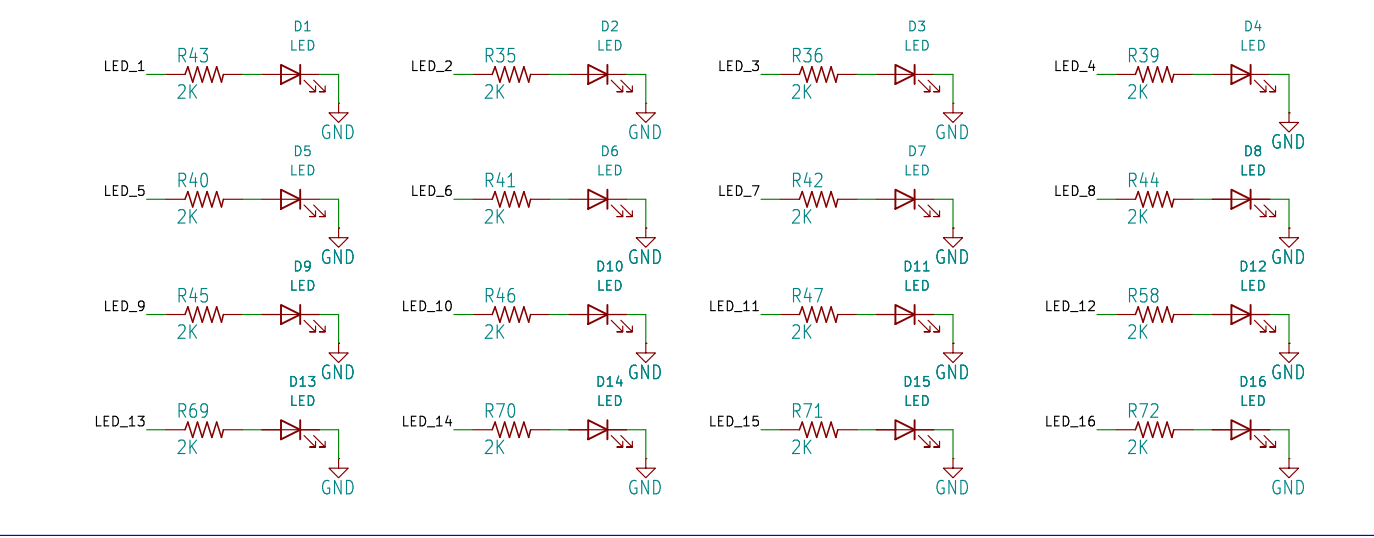
CV Input Circuit Info:
 * Cutoff = $1 / (2\pi RC) = 4.822\text{kHz}$
 * Range = $-5\text{V to } 5\text{V} \Rightarrow 3\text{V3 to } 0\text{V}$



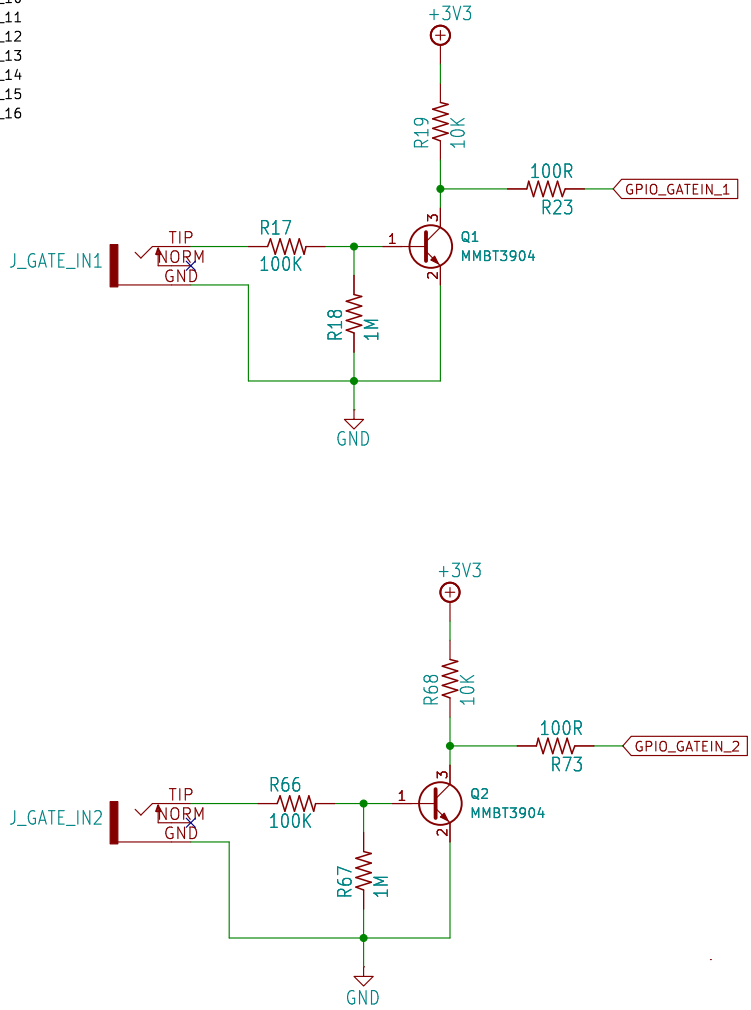
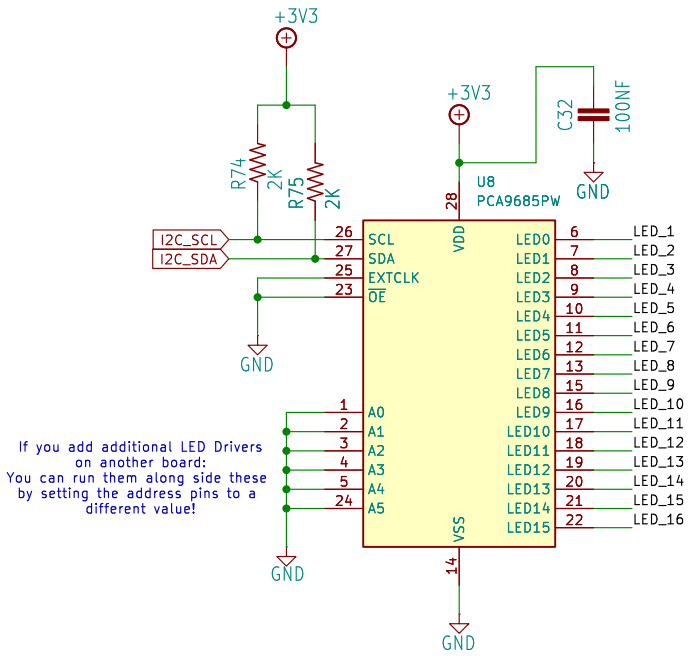
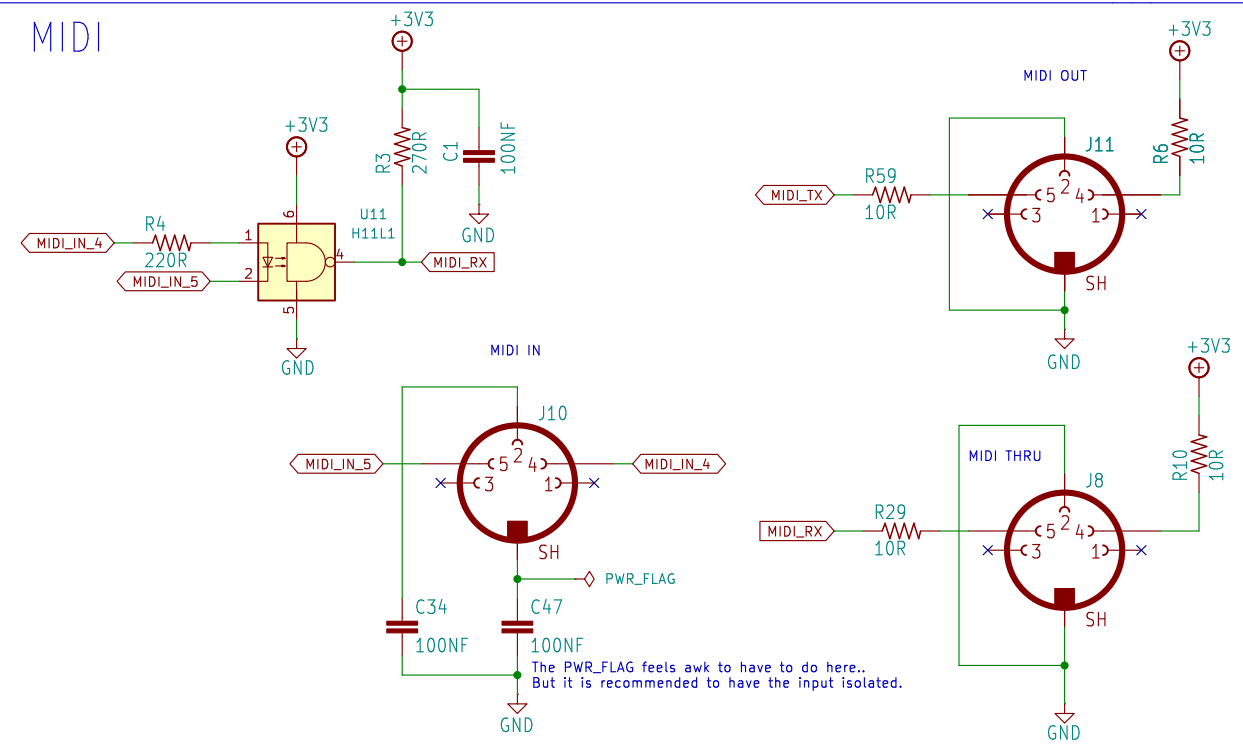
Switches + Pots



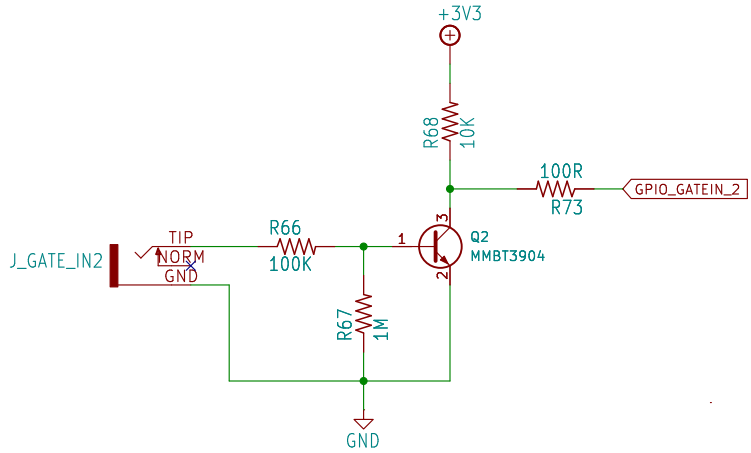
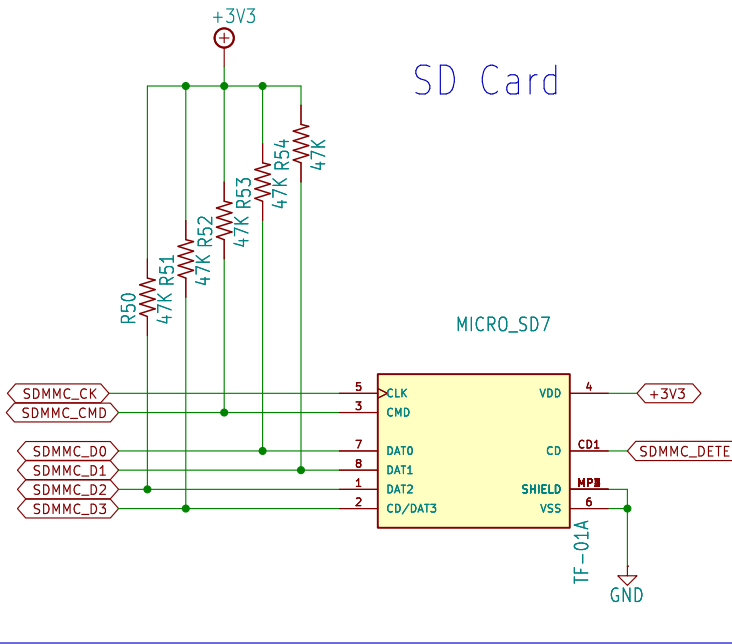
LEDs



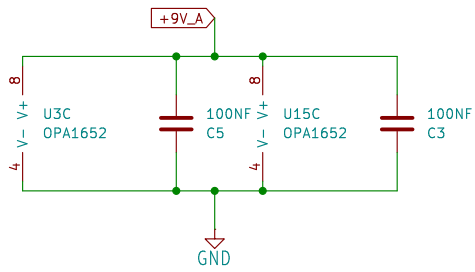
MIDI



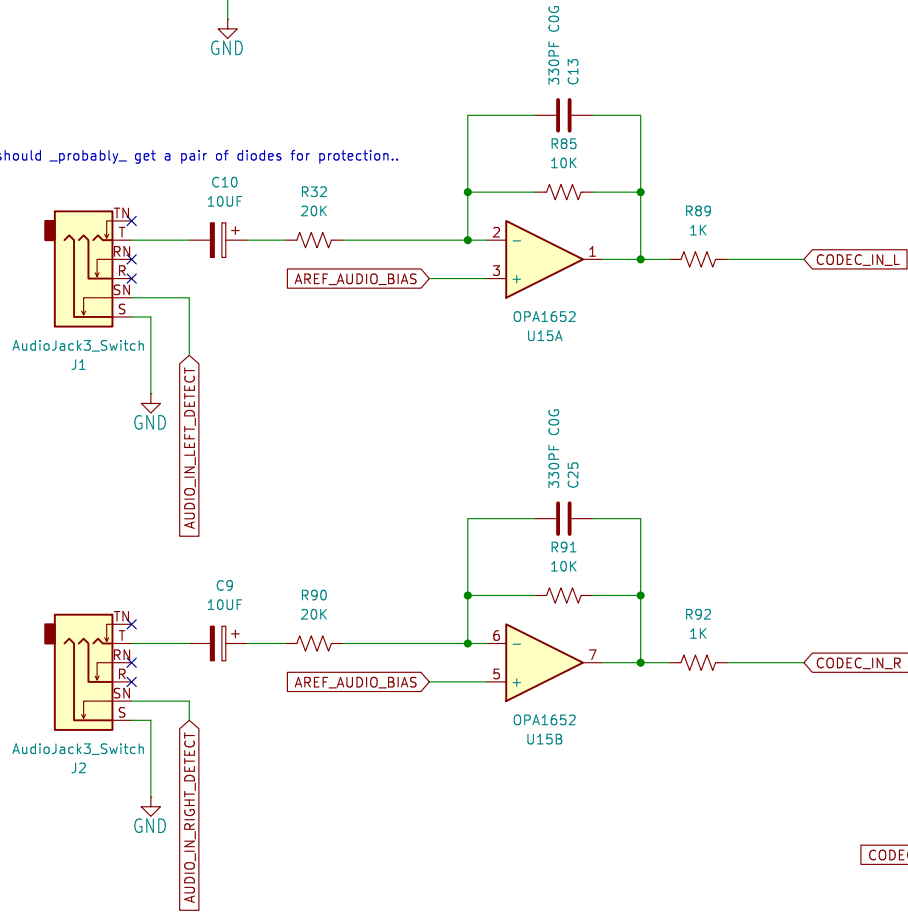
SD Card



AUDIO I/O



...s should _probably_ get a pair of diodes for protection..

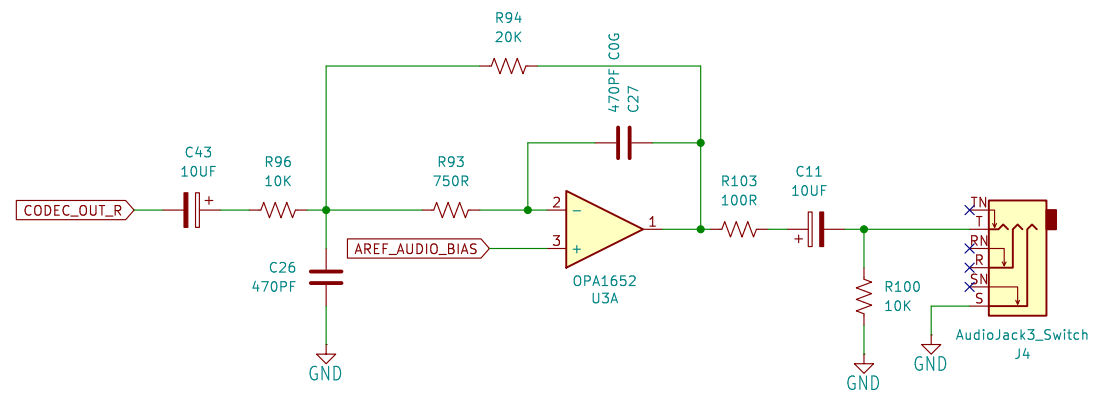


Gain is set to target +4dBu (maximum)
obviously this does not make it a pro-line-level device that way..
we'd need to probably have a certain amount of headroom.
If, instead we look at this as a -10dBV device, then we have
-14dB of headroom with these values.

$G_{in}=0.814$ ($R_{in}=10k$, $R_{fb}=16k$)
 $G_{out}=1.228$ ($R_{in}=10k$, $R_{fb}=24k$)

Alternatively we can go closer to the PCM3060's "2Vrms" reference and just do:
 $G_{in}=0.5$, $G_{out}=2$:shrug:

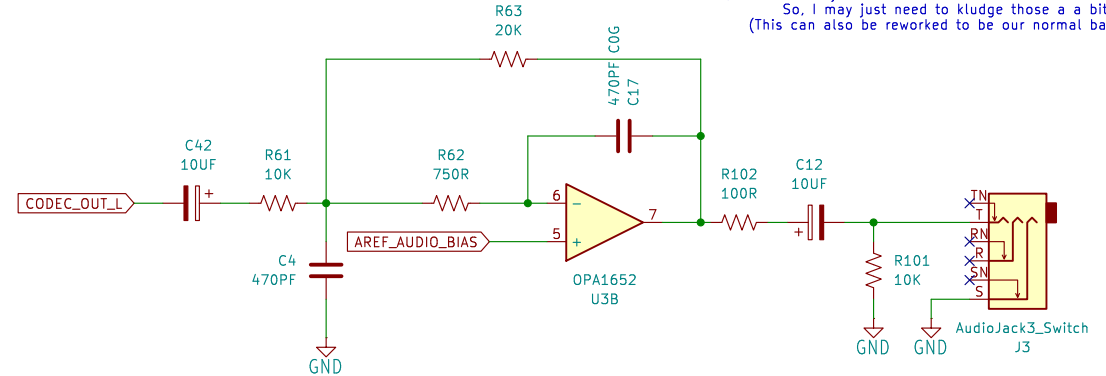
For now, I've implemented the left channel of audio with less atten/gain.



I updated these to be gain of 2.

However, I'm not really sure how the math works out for the other two values..

So, I may just need to kludge those a bit IRL.
(This can also be reworked to be our normal basic filter)



- DECISION - Two Choices:
1. Add LD1117 (or LP2985) or other 5V Linear Reg.
 2. Power TPA6110 from Seed's 3v3A

